INTERNATIONAL MONETARY FUND AND
INTERNATIONAL DEVELOPMENT ASSOCIATION

ISLAMIC REPUBLIC OF AFGHANISTAN

Joint World Bank/IMF Debt Sustainability Analysis 2007

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the International Development Association

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The external low-income country debt sustainability analysis (LIC DSA) for Afghanistan reveals a high risk of debt distress,[1,2] emphasizing the need for debt relief under the HIPC Initiative. The public debt LIC DSA suggests that, although Afghanistan's overall public sector debt dynamics remain manageable during the projection period, further fiscal adjustment (or additional grants) will be needed to ensure that domestic debt does not threaten sustainability beyond that horizon.

A. Introduction

1. The external and public debt sustainability analyses presented in this appendix are based on the common standard LIC DSA framework,[3] with some modifications to the stress tests to address data limitations in Afghanistan (Box 1). The DSAs present the projected path of Afghanistan's external and public sector debt burden indicators, and draw some conclusions on the forward-looking sustainability of debt. Methodologically, the LIC DSA

[1] These DSAs have been produced jointly by Bank and Fund staffs. Afghanistan's fiscal year runs from March 21–March 20.

[2] Afghanistan has not yet been rated under the World Bank's Country Policy and Institutional Assessment (CPIA). However, for the purposes of this analysis, the staff's have conservatively assumed a weak policy rating (and without prejudicing any future CPIA assessment). Accordingly, the corresponding indicative thresholds for the LIC DSA are 30 percent for the NPV of debt-to-GDP ratio, 100 percent for the NPV of debt-to-exports ratio, and 15 percent for the debt service-to-exports ratio. See "Operational Framework for Debt Sustainability Assessments in Low-Income Countries—Further Considerations" (http://www.imf.org/External/np/pp/eng/2005/032805.htm and IDA/R2005-0056, 3/28/05).

[3] See "Debt Sustainability in Low-Income Countries: Proposal for an Operational Framework and Policy Implications" (http://www.imf.org/external/np/pdr/sustain/2004/020304.htm and IDA/SECM2004/0035, 2/3/04) and "Debt Sustainability in Low-Income Countries: Further Considerations on an Operational Framework and Policy Implications" (http://www.imf.org/external/np/pdr/sustain/2004/091004.htm and IDA/SECM2004/0629, 9/10/04).

Box 1. Historical Data Limitations and Use of Comparator Country Data

Given the absence of data for Afghanistan prior to 2002 and continued data limitations since, the staffs did not judge it appropriate to use Afghanistan's 2–3 year of historical data to conduct the standardized alternative scenarios and stress tests. As an alternative, the analysis uses the historical averages and standard deviations of a group of comparator countries, where appropriate, in both the external and public debt templates.

The group of **comparator countries** is similar to the one used to impute the value of Afghanistan's exports of services to assess qualification under the HIPC Initiative.[1/] In this regard, the ten comparator countries: (i) are PRGF-eligible and meet the World Bank's LIC classification; and (ii) share Afghanistan's key characteristics of being landlocked and either post-conflict or located within the central Asia region. They are: Burundi, Central African Republic, Chad, Ethiopia, Kyrgyz Republic, Mongolia, Niger, Rwanda, Tajikistan, and Uzbekistan.

The **data used** was estimated based on the most recent 10-year (1997–2006) averages and standard deviations for each country and averaged across the comparator group. In this regard:

- the external debt template uses comparator country data for real GDP growth, the non-interest current account, the US dollar deflator, export value growth and net direct investment. These data are used for the alternative scenario where key variables are at their historical average (Table 2; Scenario A1), as well as the first three bound tests and the combination test (Table 2; Bound Tests B1–B3, B5); and

- the public debt template uses comparator country data on the primary balance and real GDP growth in three alternative scenarios (Table 4; Scenarios A1–A3), and the first three bound tests (Table 4; Bound Tests B1–B3).

Historical Averages & Standard Deviations, 1997-2006

	Comparator Countries		Afghanistan	
	Average	Std Dev'n	Average	Std Dev'n
Real GDP Growth	4.9	4.1	11.3	4.2
Primary Balance	-1.7	3.7	-0.9	4.0
Current Account Balance	-3.4	7.1	0.3	2.6
US$ deflator (percent change)	12.7	9.7	7.5	9.8
Value of exports of G & S, percent change	8.4	19.4	25.4	15.4

The only scenarios/tests not relying on comparator country data are: 30 percent one-time depreciation (external); the lower level of concessionality (external); and the shock to non-debt flows (external). The latter is particularly important given Afghanistan's heavy reliance on official transfers over the projection period (around 40 percent of GDP and 90 percent of official financing). However, as neither Afghanistan's historical data nor comparator country data provide a useful guide for shocks to non-debt flows, the relevant bound tests (Table 2; Bound Tests B4–B5) assume grant financing 25 percent below the baseline projection (or 12½ percent in the combined test) for 2008/09–2009/10.

1/ See Box 2 of "Islamic Republic of Afghanistan: Enhanced Heavily Indebted Poor Countries Initiative—Decision Point Document" (refer to www.imf.org).

differs from the HIPC DSA in that forward-looking debt ratios, based on single year denominators, are assessed against policy-dependent indicative thresholds. In contrast, under a HIPC DSA, debt ratios are derived on the basis of three-year backward-looking averages and assessed against uniform thresholds across countries.[4]

B. Baseline Scenario

2. **The baseline scenario relies on the same long-run macroeconomic framework as that underlying the HIPC debt relief analysis**[5] and is consistent with the program projections under the PRGF arrangement. Average annual real GDP growth is projected to moderate over the projection period, from 6.8 percent in 2006/07–2016/17 to 4.5 percent in 20017/18–2026/27. This pattern reflects the projected decrease in the investment–to-GDP ratio from short- to medium-term exceptional levels above 40 percent, to a still high yet markedly lower long-term level (around 25 percent). The latter is necessary to sustain an acceptable pace of growth in per capita consumption. Continued donor support for reconstruction and public investment is likely to be the main driving factor during the earlier years. The contribution of donor-led investment is expected to decline over the medium term with private investment gaining in prominence and eventually playing the lead role. After temporarily reaching 6.0 percent in 2007/08, inflation is projected to ease to 5 percent during the remainder of the projection period.

3. **Continued macroeconomic stability is predicated on a sustained fiscal discipline**, with the government's overall deficit (including grants) not exceeding 2½ percent of GDP over the projection period. Underlying this, revenues are assumed to increase gradually from 7 percent of GDP in 2006/07 to 12.7 percent of GDP in 2026/27, while expenditures are expected to peak at just over 22 percent of GDP in 2010/11 and stabilize below 20 percent of GDP by 2026/27.

4. **The external position is expected to remain manageable with continued donor support over the medium term**. Over the longer term, the current account deficit (excluding grants) is projected to improve gradually from over 45 percent of GDP in 2006/07 to around 10 percent in 2026/27. External grants are expected to remain the predominant form of financing,[6] but decreasing as a percentage of GDP from just under 44 percent in 2006/07 to around 7.5 percent in 2026/27. As official transfers taper off as a percent of GDP, the current account deficit (including grants) could widen over the medium term reflecting still significant development-related imports and the projected pick-up in private investment

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[4] In addition, the results of the LIC DSA differ from the HIPC DSA (main text) because of two other methodological differences related to the definition of: (i) discount rates; and (ii) exchange rates.

[5] See Box 3 of "Islamic Republic of Afghanistan: Enhanced Heavily Indebted Poor Countries Initiative—Decision Point Document" (refer to www.imf.org).

[6] Official borrowing is assumed to continue on strictly concessional terms.

(expected to be mainly financed by foreign direct investment). After averaging around 6.0 percent of GDP in 2006/07–2016/17, the external current account deficit (including grants) is anticipated to narrow to 4.6 per year in 20017/18–2026/27 (and reach 3 percent of GDP toward the end of the projection period). Income growth and improved financial intermediation are expected to stimulate private sector savings and result in a narrowing of the private sector savings-investment deficit in the long term.

5. **The baseline scenario assumes full delivery of traditional debt relief. Consistent with the baseline scenario presented for the second review under the PRGF-supported program, a financing gap is expected to be met through interim HIPC assistance as proposed for the decision point**. At this stage, the baseline does not reflect the conditional delivery of HIPC assistance at the completion point; however, this is presented in an alternative scenario.[7]

C. Debt Sustainability Analyses

External Debt Sustainability

6. **The baseline scenario, which assumes full delivery of traditional debt relief and a financing gap to be met through HIPC interim assistance, indicates that Afghanistan has a high risk of debt distress** (Table 1). Although the NPV of external debt-to-GDP ratio remains well below the threshold (30 percent), the NPV of external debt-to-exports ratio remains persistently above the 100 percent threshold, reflecting Afghanistan's very narrow export base. Also, the relatively comfortable debt service ratios (to export and revenue) are attributable to the very high degree of concessionality on the outstanding stock of debt from multilaterals (mainly on IDA or IDA-like terms).

7. **The alternative scenarios and stress tests indicate that the evolution of Afghanistan's external debt position is subject to considerable vulnerabilities** (Table 2, Figure 1). In particular, most debt indicators deteriorate significantly under the bound tests (temporary shocks) and in the face of a permanent (unfavorable) shock to the terms of new borrowing. In sharp contrast, the debt ratios improve under the historical scenario—main variables at historical average (using comparator countries) over the projection period. However, given the lack of historical data and the reliance on comparator country averages, these results should be treated cautiously. Other analyses are likely to provide more pertinent guidance in assessing Afghanistan's debt vulnerabilities.

8. **Both the scenario with less favorable terms on new borrowing and the temporary shock to non-debt creating flows underscore the vulnerabilities posed by Afghanistan's heavy reliance on official transfers and highly concessional borrowing**. A deterioration in borrowing terms by 200 basis points would delay Afghanistan's NPV of

[7] See "Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries" (refer to www.imf.org and IDA/SECM2007/0226, 03/05/2007).

external debt-to-exports ratio falling below the threshold from 2015/16 (baseline) to 2019/20. The impact of 25 percent less official transfers over 2008/09–2009/10 would be far more dramatic, with the NPV of external debt-to-exports ratio rising to 268 percent in 2009/10 and then not falling below the threshold until 2021/22.

9. **The bound tests also demonstrate that an improvement in Afghanistan's external debt indicators depends heavily on an expansion of the very narrow export base and limited export capacity**. If export growth during 2008/09 and 2009/10 were to fall to a level one standard deviation below the historical average, the NPV of external debt-to-exports would spike to 278 percent in 2009/10 and would not fall below the threshold until 2023/24. Under the combination stress test, reflecting both lower transfers and lower export growth, even the ratio of debt service-to-exports (14 percent) would approach the threshold (15 percent) in 2015/16.

10. **Financial assistance under the HIPC and Multilateral Debt Relief Initiatives (MDRI) improve significantly Afghanistan's debt situation**. Assuming the full delivery of HIPC and MDRI assistance, all three debt-burden indicators (NPV of debt-to-GDP ratio, NPV of debt-to-exports ratio, and debt service-to-exports ratio) would be significantly below the indicative thresholds.

Public Sector Debt Sustainability

11. **Under the baseline scenario, Afghanistan's public debt (including domestic debt) is expected to decline steadily as a percent of GDP until 2020/21, before increasing again** (Table 3). In the short term, the government is also running down its significant deposits at Da Afghanistan Bank (the central bank), which together with generous grant financing, explains the initial decline in the public debt ratios. However, the availability of external grants is expected to decline as a share of GDP over the projection period, from a high of just over 11 percent of GDP in 2009/10 to just under 5 percent of GDP in 2027/28. Notwithstanding the assumed steady growth of domestic revenue (excluding grants), the expected decline in grants will contribute to a pick up in the public debt ratios in the second half of the projection period.

12. **The medium- to longer-term decline in external grant financing is expected to be offset by a combination of increased revenue and domestic borrowing**. By 2027/28 domestic debt is projected to reach 11.4 percent of GDP from a zero base in 2007/08, under the expectation of gradual development in the domestic debt market. While the level of public debt over the projection period will not endanger fiscal sustainability, the average long run primary deficit, at 1.3 percent of GDP, is above the level that stabilizes the debt to GDP ratio.

13. **Debt dynamics remain manageable under the various stress tests** (Table 4). Although the one standard deviation shock to GDP doubles the NPV of debt-to-GDP ratio to 25 percent, relative to the baseline, this remains below the relevant country specific debt

burden thresholds for countries rated as weak performers against the World Bank's CPIA. Baseline projections for the evolution of debt ratios are not seen to be overly optimistic as they do not differ significantly from historical trends.[8]

D. Debt Distress Classification and Conclusion

14. **Afghanistan's risk of debt distress is high**.

- The baseline path of NPV of external debt-to-exports remains above the LIC DSA threshold (100 percent) until 2015/16. The sensitivity analysis indicates that Afghanistan's external debt sustainability is heavily dependant on the continued availability of mainly official grant financing, supplemented by highly concessional loans.

- The emerging role of domestic debt financing over the longer term does not pose an immediate risk to the debt outlook, but it does underscore the need for cautious approach to fiscal policy as the government builds the domestic revenue base and given uncertainties about the pace of donor expenditures being brought on budget. While the level of public debt remains manageable over the projection period, the government will need additional grant financing and/or fiscal consolidation to contain the debt dynamics beyond that horizon.

[8] Care should be taken when making this comparison on account of the methodology used to overcome the lack of historical data in Afghanistan.

Table 1. Islamic Republic of Afghanistan: External Debt Sustainability Framework, Baseline Scenario, 2004/05–2027/28 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical	Standard	Projections									
	2004/05	2005/06	2006/07	Average 6/	Deviation 6/	2007/08	2008/09	2009/10	2010/11	2011/12	2012/13	2007/08-2012/13 Average	2017/18	2027/28	2017/18-2027/28 Average
External debt (nominal) 1/	**10.9**	**162.2**	**20.0**			**19.4**	**18.7**	**17.8**	**17.3**	**16.8**	**16.2**		**14.6**	**7.2**	
o/w public and publicly guaranteed (PPG)	10.9	162.2	20.0			19.4	18.7	17.8	17.3	16.8	16.2		14.6	7.2	
Change in external debt	-3.0	151.2	-142.2			-0.6	-0.7	-0.9	-0.5	-0.6	-0.5		-0.3	-0.7	
Identified net debt-creating flows	-8.1	-6.3	-20.2			-2.8	-2.0	-1.3	0.2	1.1	1.3		0.3	-0.5	
Non-interest current account deficit	**-1.9**	**-0.7**	**0.5**	**-0.3**	**2.6**	**2.0**	**3.1**	**4.5**	**5.6**	**6.3**	**6.9**		**6.0**	**2.6**	4.8
Deficit in balance of goods and services	45.0	42.1	44.6			40.9	39.1	37.6	34.9	32.0	29.5		19.9	10.0	
Exports	9.0	9.0	9.6			9.7	9.8	10.0	10.1	10.2	10.4		11.7	13.1	
Imports	54.0	51.2	54.2			50.6	48.9	47.5	45.0	42.2	39.9		31.7	23.1	
Net current transfers (negative = inflow)	-46.7	-42.5	-43.4	-43.3	8.7	-38.1	-35.1	-32.2	-28.5	-24.9	-21.9		-13.5	-7.2	-11.7
o/w official	-41.5	-37.3	-38.2			-33.1	-30.2	-27.4	-23.9	-20.4	-17.6		-10.0	-5.5	
Other current account flows (negative = net inflow)	-0.2	-0.3	-0.7			-0.9	-0.9	-0.9	-0.8	-0.8	-0.7		-0.4	-0.3	
Net FDI (negative = inflow)	**-3.1**	**-3.7**	**-2.8**	**-2.4**	**1.1**	**-3.1**	**-4.1**	**-5.1**	**-4.8**	**-4.7**	**-5.1**		**-5.4**	**-2.8**	-4.5
Endogenous debt dynamics 2/	**-3.2**	**-1.9**	**-18.0**			**-1.7**	**-0.9**	**-0.7**	**-0.5**	**-0.5**	**-0.4**		**-0.3**	**-0.2**	
Contribution from nominal interest rate	0.1	0.1	0.6			0.6	0.5	0.4	0.4	0.4	0.4		0.3	0.1	
Contribution from real GDP growth	-0.9	-1.2	-10.7			-2.2	-1.4	-1.1	-1.0	-0.9	-0.9		-0.6	-0.3	
Contribution from price and exchange rate changes	-2.4	-0.7	-7.9			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**5.1**	**157.5**	**-122.0**			**2.2**	**1.2**	**0.4**	**-0.7**	**-1.7**	**-1.8**		**-0.7**	**-0.2**	
o/w exceptional financing	-0.3	-0.4	-128.4			-0.5	-0.4	-0.3	-0.2	-0.1	-0.1		-0.2	-0.1	
NPV of external debt 4/	...	...	16.0			14.7	14.0	13.3	12.8	12.3	11.7		9.9	5.1	
In percent of exports	...	...	166.6			152.2	142.1	133.1	126.7	120.0	113.1		84.3	39.1	
NPV of PPG external debt	**...**	**...**	**16.0**			**14.7**	**14.0**	**13.3**	**12.8**	**12.3**	**11.7**		**9.9**	**5.1**	
In percent of exports	**...**	**...**	**166.6**			**152.2**	**142.1**	**133.1**	**126.7**	**120.0**	**113.1**		**84.3**	**39.1**	
In percent of government revenues	**...**	**...**	**300.9**			**201.0**	**178.3**	**157.4**	**141.2**	**128.1**	**115.2**		**81.3**	**40.3**	
Debt service-to-exports ratio (in percent)	**1.4**	**1.4**	**7.1**			**6.1**	**5.3**	**4.8**	**4.5**	**4.7**	**5.6**		**4.7**	**3.9**	
PPG debt service-to-exports ratio (in percent)	**1.4**	**1.4**	**7.1**			**6.1**	**5.3**	**4.8**	**4.5**	**4.7**	**5.6**		**4.7**	**3.9**	
PPG debt service-to-revenue ratio (in percent)	**2.9**	**2.2**	**12.8**			**8.1**	**6.7**	**5.7**	**5.0**	**5.1**	**5.7**		**4.5**	**4.0**	
Total gross financing need (billions of U.S. dollars)	-0.3	-0.3	-0.1			-0.1	-0.1	0.0	0.2	0.3	0.4		0.3	0.2	
Non-interest current account deficit that stabilizes debt ratio	1.2	-151.9	142.7			2.5	3.8	5.4	6.1	6.9	7.4		6.4	3.3	
Key macroeconomic assumptions															
Real GDP growth (in percent)	8.0	14.0	7.5	11.3	4.2	13.0	8.4	6.6	6.1	6.0	5.8	7.6	4.5	4.5	4.7
GDP deflator in US dollar terms (change in percent)	20.6	7.4	5.1	7.5	9.8	4.2	4.7	4.9	4.9	5.0	5.0	4.8	5.0	5.0	5.0
Effective interest rate (percent) 5/	0.7	0.9	0.4	0.7	0.2	3.3	2.9	2.7	2.7	2.8	2.9	2.9	2.2	1.3	2.0
Growth of exports of G&S (US dollar terms, in percent)	11.1	23.5	19.8	25.4	15.4	18.9	14.9	13.5	12.8	12.8	12.6	14.3	14.3	9.1	11.7
Growth of imports of G&S (US dollar terms, in percent)	14.6	16.0	19.6	29.2	25.0	10.0	9.7	8.6	5.4	4.4	4.9	7.2	5.7	6.8	6.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	46.7	48.7	51.1	50.9	49.4	48.4	49.2	42.9	32.4	39.5
Aid flows (in billions of US dollars) 7/	2.8	2.9	13.9			3.5	3.6	3.6	3.5	3.3	3.2		3.2	4.3	
o/w Grants	2.5	2.7	3.2			3.2	3.3	3.4	3.3	3.1	3.0		2.8	3.9	
o/w Concessional loans	0.3	0.1	0.2			0.2	0.2	0.2	0.2	0.2	0.2		0.4	0.3	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			34.0	31.0	28.1	24.6	21.1	18.2		10.6	5.6	9.1
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			97.0	97.4	97.8	97.1	96.8	96.3		93.7	95.2	94.7
Memorandum items:															
Nominal GDP (billions of US dollars)	6.0	7.3	8.3			9.7	11.0	12.3	13.7	15.3	17.0		27.9	70.5	
(NPVt-NPVt-1)/GDPt-1 (in percent)						1.4	1.1	0.9	1.0	0.9	0.8	1.0	0.7	0.0	0.4

Source: Staff simulations.

1/ Includes both public and private sector external debt. Afghanistan's fiscal year runs from March 21 to March 20.
2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the NPV of new debt).

Table 2. Islamic Republic of Afghanistan: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007/08–27/28

(In percent)

	Projections 1/							
	2007/08	2008/09	2009/10	2010/11	2011/12	2012/13	2017/18	2027/28
NPV of debt-to-GDP ratio								
Baseline	15	14	13	13	12	12	10	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008/09-27/28 2/	15	15	14	13	11	10	6	6
A2. New public sector loans on less favorable terms in 2008/09-27/28 3/	15	14	14	14	14	13	13	8
A3. Full delivery of HIPC assistance & MDRI	7	7	8	8	8	7	8	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008/09-09/10	15	15	15	15	14	13	11	6
B2. Export value growth at historical average minus one standard deviation in 2008/09-09/10 4/	15	15	17	16	15	15	12	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	15	14	14	13	13	12	10	5
B4. Net non-debt creating flows 25 percent below the baseline in 2008/09-09/10 5/	15	21	27	25	24	23	17	7
B5. Combination of B1-B4 using one-half standard deviation shocks	15	20	27	25	24	23	17	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008/09 6/	15	20	19	18	17	16	14	7
NPV of debt-to-exports ratio								
Baseline	152	142	133	127	120	113	84	39
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008/09-27/28 2/	152	149	142	129	112	95	51	44
A2. New public sector loans on less favorable terms in 2008/09-27/28 3/	152	146	140	136	132	128	107	60
A3. Full delivery of HIPC assistance & MDRI	74	76	75	75	74	72	64	38
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008/09-09/10	152	142	133	127	120	113	84	39
B2. Export value growth at historical average minus one standard deviation in 2008/09-09/10 4/	152	201	278	263	248	233	164	71
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	152	142	133	127	120	113	84	39
B4. Net non-debt creating flows 25 percent below the baseline in 2008/09-09/10 5/	152	212	268	251	235	219	142	54
B5. Combination of B1-B4 using one-half standard deviation shocks	152	221	316	297	278	260	172	68
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008/09 6/	152	142	133	127	120	113	84	39
Debt service-to-exports ratio								
Baseline	6	5	5	5	5	6	5	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008/09-27/28 2/	6	5	5	4	4	4	3	2
A2. New public sector loans on less favorable terms in 2008/09-27/28 3/	6	5	5	5	5	6	6	5
A3. Full delivery of HIPC assistance & MDRI	0	0	1	2	2	2	2	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008/09-09/10	6	5	5	5	5	6	5	4
B2. Export value growth at historical average minus one standard deviation in 2008/09-09/10 4/	6	7	8	9	9	10	10	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	6	5	5	5	5	6	5	4
B4. Net non-debt creating flows 25 percent below the baseline in 2008/09-09/10 5/	6	5	6	7	7	8	9	6
B5. Combination of B1-B4 using one-half standard deviation shocks	6	6	8	9	9	10	11	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008/09 6/	6	5	5	5	5	6	5	4
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 7/	41	41	41	41	41	41	41	41

Source: Staff projections and simulations.

1/ Afghanistan's fiscal year runs from March 21 to March 20.
2/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows. Given that Afghanistan has only limited historical data, the historical average scenario was based on 10-year averages of comparator countries (see Box B1).
3/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.
4/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
5/ Includes official and private transfers and FDI.
6/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
7/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Islamic Republic of Afghanistan: Public Sector Debt Sustainability Framework, Baseline Scenario, 2004/05–2027/28

(In percent of GDP, unless otherwise indicated)

	Actual					Est.											Projections											
	2004/05	2005/06	2006/07	Historical Average 5/	Standard Deviation 5/	2007/08	2008/09	2009/10	2010/11	2011/12	2012/13	2013/14	2014/15	2015/16	2016/17	Average 2007/08-2016/17	2017/18	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	2027/28	Average 2017/18-2027/28
Public sector debt 1/	11.4	162.8	20.0			19.4	18.7	17.8	17.3	16.8	16.2	15.8	16.1	16.1	16.1		16.0	15.9	15.7	15.6	15.8	16.1	16.5	17.1	17.6	18.1	18.6	
o/w foreign-currency denominated	11.4	162.8	20.0			19.4	18.7	17.8	17.3	16.8	16.2	15.8	15.5	15.2	14.9		14.6	14.0	13.2	12.4	11.6	10.8	10.1	9.3	8.6	7.9	7.2	
Change in public sector debt	-2.7	151.3	-142.8			-0.6	-0.7	-0.9	-0.5	-0.6	-0.5	-0.4	0.3	0.0	-0.1		-0.1	-0.2	-0.2	-0.1	0.1	0.3	0.4	0.6	0.5	0.5	0.5	
Identified debt-creating flows	-2.3	-3.5	-144.8			-0.7	-1.2	-1.4	0.1	0.7	0.8	0.8	0.5	0.2	0.0		0.0	-0.1	-0.2	-0.1	0.1	0.3	0.4	0.6	0.5	0.5	0.5	
Primary deficit	1.2	-1.0	2.6	-0.9	4.0	2.5	1.9	2.1	1.9	2.1	2.1	2.1	1.7	1.4	1.2	2.1	1.1	0.9	0.9	0.9	1.1	1.2	1.3	1.4	1.4	1.3	1.3	1.3
Revenue and grants	12.7	15.6	15.7			17.4	18.9	19.5	19.7	19.3	19.1	18.9	19.0	19.1	19.1		19.1	19.1	19.0	18.9	18.6	18.3	18.0	17.7	17.6	17.5	17.5	
of which : grants	8.2	9.9	8.7			10.1	11.0	11.1	10.7	9.7	8.9	8.2	7.8	7.5	7.2		6.9	6.7	6.4	6.1	5.9	5.6	5.3	5.1	4.9	4.9	4.8	
Primary (noninterest) expenditure	13.8	14.6	18.3			20.0	20.8	21.6	21.7	21.4	21.2	21.0	20.7	20.4	20.3		20.2	20.0	19.9	19.8	19.6	19.5	19.3	19.2	19.0	18.8	18.7	
Automatic debt dynamics	-3.1	-2.2	-18.5			-2.6	-1.9	-1.6	-1.5	-1.4	-1.4	-1.3	-1.2	-1.2	-1.2		-1.1	-1.1	-1.0	-1.0	-0.9	-0.9	-0.9	-0.8	-0.8	-0.8	-0.8	
Contribution from interest rate/growth differential	-2.9	-2.4	-18.9			-2.6	-1.9	-1.6	-1.5	-1.4	-1.4	-1.3	-1.2	-1.2	-1.2		-1.1	-1.1	-1.0	-1.0	-0.9	-0.9	-0.9	-0.8	-0.8	-0.8	-0.8	
of which : contribution from average real interest rate	-1.8	-1.0	-7.6			-0.3	-0.4	-0.4	-0.4	-0.5	-0.4	-0.4	-0.4	-0.4	-0.4		-0.4	-0.4	-0.3	-0.3	-0.3	-0.2	-0.2	-0.1	-0.1	0.0	0.0	
of which : contribution from real GDP growth	-1.0	-1.4	-11.3			-2.3	-1.5	-1.2	-1.0	-1.0	-0.9	-0.8	-0.8	-0.8	-0.8		-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.8	-0.8	
Contribution from real exchange rate depreciation	-0.3	0.2	0.4			0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		...	...	...	...	...	...	...	...	...	...	...	
Other identified debt-creating flows	-0.3	-0.4	-128.9			-0.6	-1.2	-1.9	-0.4	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			-0.2	-0.9	-1.6	-0.4	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Debt relief (HIPC and other)	-0.3	-0.4	-128.9			-0.4	-0.3	-0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes	-0.5	154.9	2.0			0.1	0.5	0.5	-0.6	-1.3	-1.3	-1.2	-0.2	-0.1	-0.1		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
NPV of public sector debt	0.0	15.4	13.6			11.6	10.3	9.3	8.4	7.6	6.9	6.2	6.2	5.9	5.6		5.4	5.4	5.6	5.9	6.5	7.3	8.1	9.1	10.1	11.1	12.1	
o/w foreign-currency denominated	0.0	15.4	13.6			11.6	10.3	9.3	8.4	7.6	6.9	6.2	5.5	4.9	4.4		3.9	3.5	3.1	2.7	2.4	2.0	1.7	1.4	1.1	0.9	0.7	
o/w external	...	15.4	13.6			11.6	10.3	9.3	8.4	7.6	6.9	6.2	5.5	4.9	4.4		3.9	3.5	3.1	2.7	2.4	2.0	1.7	1.4	1.1	0.9	0.7	
NPV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...	...	...	...	...		...	...	...	...	...	...	...	...	...	...	...	
Gross financing need 2/	1.3	-0.8	3.2			3.0	2.4	2.5	2.3	2.5	2.5	2.5	2.2	2.5	2.6		2.7	2.9	3.2	3.9	4.8	5.9	7.1	8.3	9.5	10.6	11.8	
NPV of public sector debt-to-revenue ratio (in percent) 3/	0.0	98.6	87.0			66.8	54.8	47.6	42.6	39.4	35.9	32.6	32.5	30.7	29.3		28.2	28.0	29.2	31.3	35.1	40.0	45.2	51.5	57.4	63.3	69.2	
o/w external	...	98.6	87.0			66.8	54.8	47.6	42.6	39.4	35.9	32.6	29.1	25.9	23.0		20.6	18.2	16.2	14.3	12.7	11.0	9.4	7.9	6.3	5.2	4.2	
Debt service-to-revenue ratio (in percent) 3/ 4/	1.1	0.8	3.7			2.9	2.4	2.1	1.9	2.0	2.0	2.2	2.4	2.7	2.9		3.0	3.2	3.4	3.8	4.2	4.7	5.2	5.8	6.3	6.7	7.1	
Primary deficit that stabilizes the debt-to-GDP ratio	3.9	-152.3	145.4			3.1	2.6	3.0	2.4	2.7	2.7	2.5	1.4	1.3	1.3		1.1	1.1	1.0	1.0	0.9	0.9	0.9	0.8	0.8	0.8	0.8	
Key macroeconomic and fiscal assumptions																												
Real GDP growth (in percent)	8.0	14.0	7.5	9.8	3.6	13.0	8.4	6.6	6.1	6.0	5.8	5.5	5.3	5.2	5.2	7.6	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.7
Average nominal interest rate on forex debt (in percent)	0.7	0.9	0.4	0.7	0.2	2.7	2.4	2.3	2.2	2.1	2.1	2.1	2.0	2.0	1.9	2.3	1.9	1.8	1.8	1.7	1.7	1.7	1.6	1.6	1.6	1.5	1.4	1.8
Average real interest rate on domestic currency debt (in percent)	...	...	...	...	...	...	...	...	...	...	...	...	...	3.7	3.3	...	3.1	3.3	3.4	3.3	3.3	3.2	3.0	3.0	2.8	2.7	2.7	3.1
Real exchange rate depreciation (in percent, + indicates depreciation	-2.3	2.5	0.3	-1.2	3.4	0.0	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	17.0	11.9	5.7	10.3	5.3	4.5	4.7	4.9	4.9	5.0	5.0	5.0	5.0	5.0	5.0	4.9	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, in percent	7.0	20.8	34.2	20.6	13.6	23.5	12.6	11.1	6.1	4.8	4.8	4.2	4.0	3.9	4.3	10.5	4.0	3.9	3.9	3.8	3.8	3.7	3.6	3.6	3.6	3.6	4.1	3.9
Grant element of new external borrowing (in percent)	...	...	66.9	66.9	...	66.9	66.9	66.9	63.8	62.4	61.4	60.2	58.5	56.1	56.1	64.7	56.1	56.4	54.4	54.4	51.2	51.2	48.6	48.6	48.6	46.0	46.0	...

Sources: Afghan authorities; and Fund staff estimates and projections.

1/ Fiscal reporting covers only central government. Debt is net debt.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 3 years, subject to data availability.For real GDP growth and the primary deficit, averages and standard deviations are calculated from comparator countries.

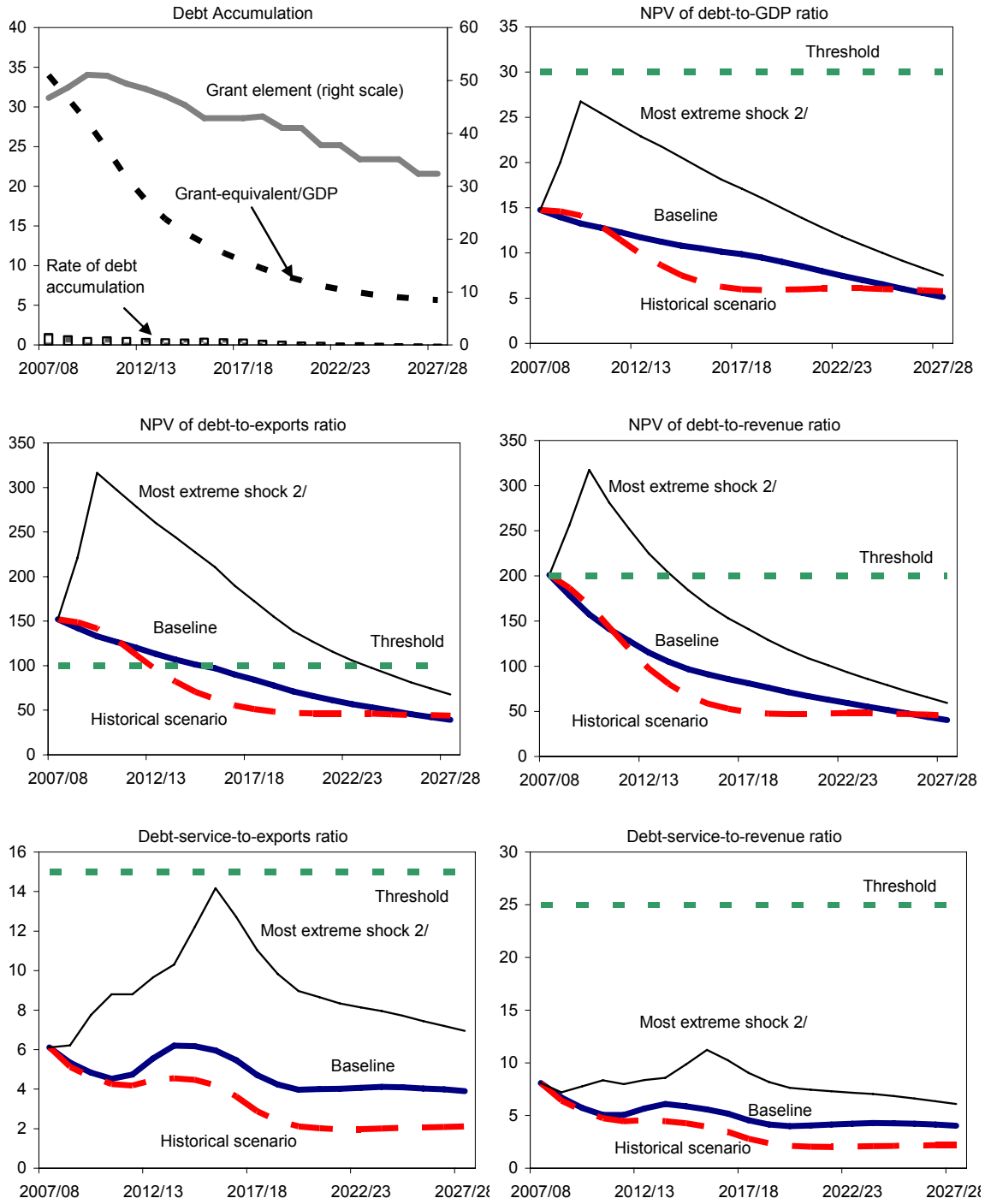
Table 4. Islamic Republic of Afghanistan: Sensitivity Analysis for Key Indicators of Public Debt 2007/08–2027/28

	Projections							
	2007/08	2008/09	2009/10	2010/11	2011/12	2012/13	2017/18	2027/28
NPV of Debt-to-GDP Ratio								
Baseline	12	10	9	8	8	7	5	12
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	12	11	9	8	7	6	6	15
A2. Primary balance is unchanged from 2007/08	12	11	10	9	9	8	9	20
A3. Permanently lower GDP growth 1/	12	10	10	9	8	8	9	25
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008/09-2009/10	12	11	12	11	11	11	12	25
B2. Primary balance is at historical average minus one standard deviations in 2008/09-2009/10	12	13	13	12	11	10	8	13
B3. Combination of B1-B2 using one half standard deviation shocks	12	12	12	10	8	7	3	9
B4. 10 percent of GDP increase in other debt-creating flows in 2008/09	12	17	15	14	12	11	8	13
NPV of Debt-to-Revenue Ratio 2/								
Baseline	67	55	48	43	39	36	28	69
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	67	55	46	41	36	32	28	83
A2. Primary balance is unchanged from 2007/08	67	57	51	47	45	43	48	115
A3. Permanently lower GDP growth 1/	67	55	49	44	42	40	44	136
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008/09-2009/10	67	58	55	53	53	53	62	137
B2. Primary balance is at historical average minus one standard deviations in 2008/09-2009/10	67	66	68	61	57	53	40	75
B3. Combination of B1-B2 using one half standard deviation shocks	67	61	58	48	42	35	17	51
B4. 10 percent of GDP increase in other debt-creating flows in 2008/09	67	88	76	68	64	59	44	77
Debt Service-to-Revenue Ratio 2/								
Baseline	3	2	2	2	2	2	3	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	3	2	2	2	2	2	3	8
A2. Primary balance is unchanged from 2007/08	3	2	2	2	2	2	4	9
A3. Permanently lower GDP growth 1/	3	2	2	2	2	2	4	10
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008/09-2009/10	3	2	2	2	2	2	4	11
B2. Primary balance is at historical average minus one standard deviations in 2008/09-2009/10	3	2	2	2	2	2	4	8
B3. Combination of B1-B2 using one half standard deviation shocks	3	2	2	2	2	2	3	6
B4. 10 percent of GDP increase in other debt-creating flows in 2008/09	3	2	3	2	2	2	4	8

Sources: Afghan authorities; and Fund staff estimates and projections.

1/ Real GDP growth is assumed to be the average of the comparator group minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

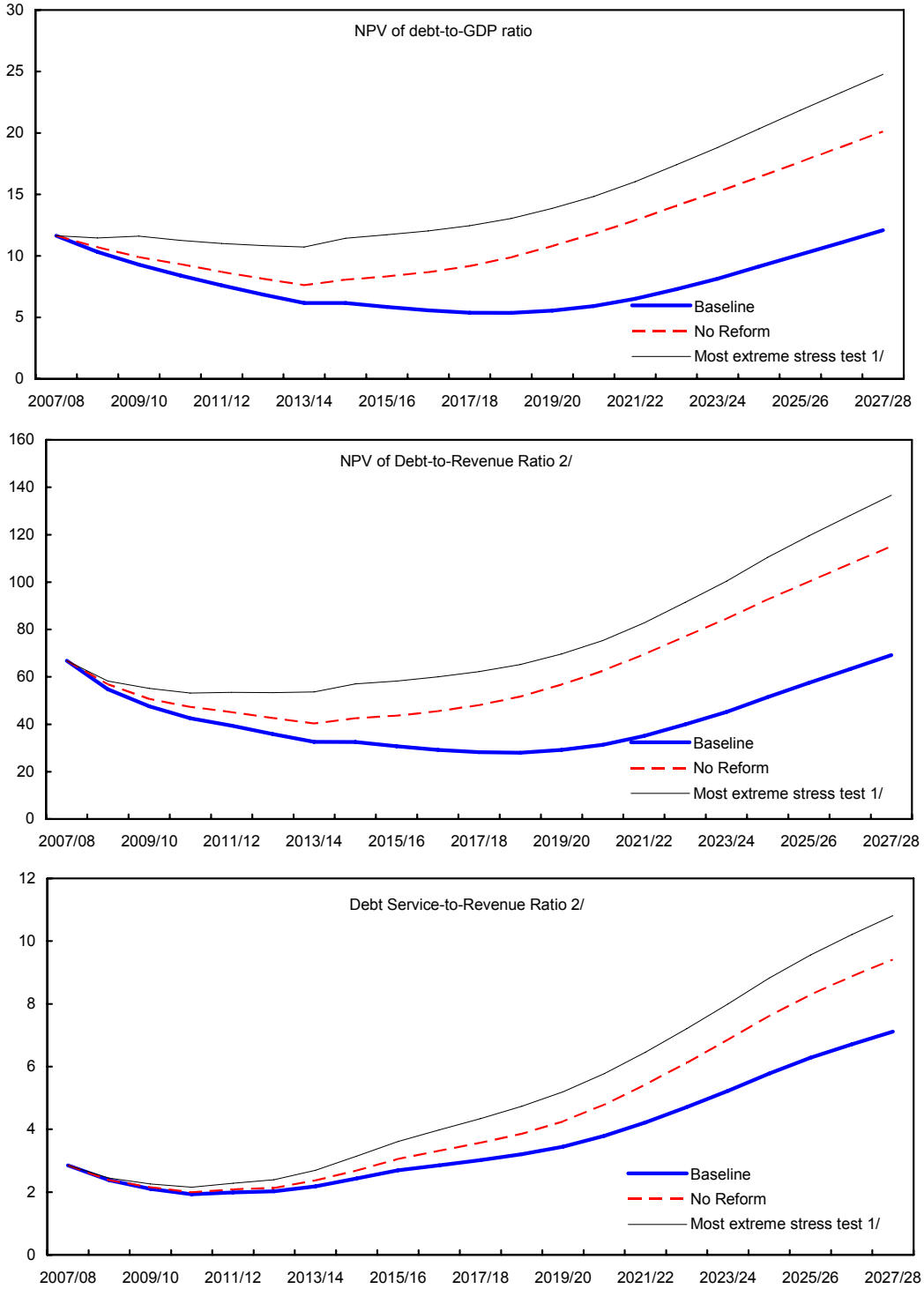
Figure 1. Islamic Republic of Afghanistan: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2007/08–2027/28 1/



Sources: Staff projections and simulations.

1/ Afghanistan's fiscal year runs from March 21 to March 20.
2/ Combination shock B5 (see Table 2), using one-half standard deviation shocks.

Figure 2. Islamic Republic of Afghanistan: Indicators of Public Debt Under Alternative Scenarios, 2007/08–2027/28



Source: Staff projections and simulations.

1/ Most extreme stress test is that which yields highest ratio in 2017/18. This resulted from the stress test under which real GDP growth in 2008/09 and 2009/10 was at historical average minus one standard deviation (based on comparator country data).
2/ Revenue including grants.